**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# Client-Server and Peer-to-Peer Networks

**Name** _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

## Do Now - Four retrieval questions

BLACK PEN: answer four questions before checking the answers.

1. What distinguishes a LAN?

2. How is an item accessed in a 1D array?

A) Using its font
B) Using an index
C) Using a file extension
D) Using a network address

3. What is the difference between input and output?

4. A ticket program checks ages. "A few successful tests prove it has no faults." Correct this. How should it be tested?

## Today's objectives

Explain what a client and a server do. Describe peer-to-peer and client-server arrangements.

## Key vocabulary - Read, cover, recall

**Client**
Device that requests a network service.

**Server**
Device that provides a service or resource.

**Client-server**
Network with clients and separate service-providing servers.

**Peer-to-peer**
Devices directly sharing resources as peers.

**Write one key term and its meaning in your own words:**

| Do Now | I Do | We Do | You Do | Review & Improve |
|---|---|---|---|---|
| **Recall previous learning** | Read new knowledge | Practise with guidance | Work independently | Check my understanding |

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# Learn and practise

## New learning - Key ideas

| | |
|---|---|
| **Client-server** | Clients ask for services; servers provide them. The two computers have different roles. |
| **Peer-to-peer** | Peers can each share and request files without needing a dedicated central server. |
| **Client-server benefits** | Central permissions, user management and backups can help larger organisations; dedicated servers need investment. |
| **Peer-to-peer trade-offs** | A small group can share directly at low cost, but there is no central backup and files may be unavailable if the sharing device is switched off. |

## I Do - One worked example

**A secondary school needs user logins, shared files and backups. Explain TWO advantages of client-server networking. [4 marks]**

1. The school needs controlled accounts and reliable shared files.

2. A central server can manage logins/permissions [1], making access control easier [1].

3. Files are backed up centrally [1], helping restore data after failure [1].

4. Two separate advantages, each supported by a consequence, gain four marks.

## We Do - Guided questions

BLACK PEN: use the model, then explain your answer.

1. A small art group with four laptops shares files directly. Explain why peer-to-peer could be suitable. [4 marks]

2. A health clinic stores confidential files. Explain two advantages of a managed client-server network. [4 marks]

3. A peer shares a file but switches off their computer. Explain one disadvantage of peer-to-peer and another related to security. [4 marks]

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# Think and write

## Code First - Predict > Write > Test > Improve

**YOUR TASK: Predict the message.**

```
permission = true
if permission then
print("Access granted")
else
print("Access denied")
endif
```

BLACK PEN: predict or complete the code; test, then improve.

For extended answers, continue on lined paper if needed.

## You Do - Independent exam questions

Q2. State the role of a server. [1 mark]

Q3. A community drama group directly shares sound files between its rehearsal laptop and another computer, without a dedicated server. Explain one benefit and one drawback of peer-to-peer networking. [4 marks]

Q4. A leisure centre has over 100 staff accounts and shared booking records. Compare client-server and peer-to-peer networking for this organisation. [6 marks]

## Exit Ticket - True or False?

Circle T or F before reviewing Page 4.

1. All peer-to-peer devices must request files from one dedicated central server. ☐ **T** ☐ **F**

2. Clients ask for services; servers provide them ☐ **T** ☐ **F**

3. Peers can each share and request files without needing a dedicated central server. ☐ **T** ☐ **F**

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# Review & improve

GREEN PEN: tick correct marking points, then correct errors.

Choose your hardest question to rewrite in full below.

## Do Now - Answers

**Q1** It connects devices across a small area, such as one school site. ☐

**Q2** B - Using an index ☐

**Q3** Input enters a program; output is information the program produces. ☐

**Q4** Test valid and invalid inputs. Compare actual and expected results to find faults. A few successful tests cannot prove there are no faults. ☐

## Code First - Model

```
permission = true
if permission then
print("Access granted")
else
print("Access denied")
endif
```

**Output: Access granted**

## We Do - Check the model

1. It needs no dedicated server [1], keeping setup cheap [1]. Laptops share files directly [1], which is simple for a small group [1]. ☐

2. Central permissions restrict access [1], supporting confidentiality [1]. Central backups [1] let records be restored after a failure [1]. ☐

3. The resource becomes unavailable [1], interrupting other users [1]. Without central security management [1], controlling access consistently becomes harder [1]. ☐

## Exit Ticket - Check

**1. FALSE 2. TRUE 3. TRUE**

## You Do - Self-mark

**Q2** Provides services/resources to clients [1]. ☐

**Q3** No dedicated server needed [1], reducing cost [1]. The other laptop can be off [1], preventing shared-file access [1]. ☐

**Q4** Client-server: central server provides services [1], enabling permissions [1] and central backups [1]. Peer-to-peer: devices share directly [1] with cheap setup [1], but central control and backups are harder or files may be unavailable when peers are off [1]. ☐

## Try the question again

**1. Copy the FULL question you found most difficult:**

**2. In GREEN PEN, write a complete new answer:**